

a lightweight messaging application. Chatbots can be classified into the following two categories.

- **Rule based chatbots:** These work like interactive FAQs. They are programmed to recognise certain terms and patterns, from which they can respond to the end user using pre-set answers.
- **AI chatbots:** These chatbots act as artificial brains and use sophisticated cognitive and natural language processing capabilities. AI chatbots can understand requests along with context, intent and emotion, making use of machine learning and deep learning techniques to become smarter, and conduct more elaborate, real-time conversations.

The process of creating a chatbot

Similar to the process of designing any software or website, developing chatbot based software involves following a series of steps. Figure 1 highlights the stages of chatbot development.

Stage 1 - Plan: It is very important to have an in-depth understanding of the goals, processes and user requirements of the chatbot before starting development. It is also important to review all the design guidelines for stability and to check if they meet the requirements of the chatbot. The bot should be designed with speech and natural language processing capabilities, in order to offer Q/A solutions to end users.

Stage 2 - Build: Chatbots bring in a conversational interface for the platform and end users to interact, and to send or receive messages. The following functionalities should be added to the chatbot to make it a complete build.

- **Natural language processing** makes the bot understand language, detect spelling mistakes and identify the users' requirements.
- **Answer-processing:** The chatbot needs a strong knowledge base to handle the real-time queries of the user in a natural way.
- **Manage multiple models:** The



Figure 1: Chatbot development (Source: Microsoft)

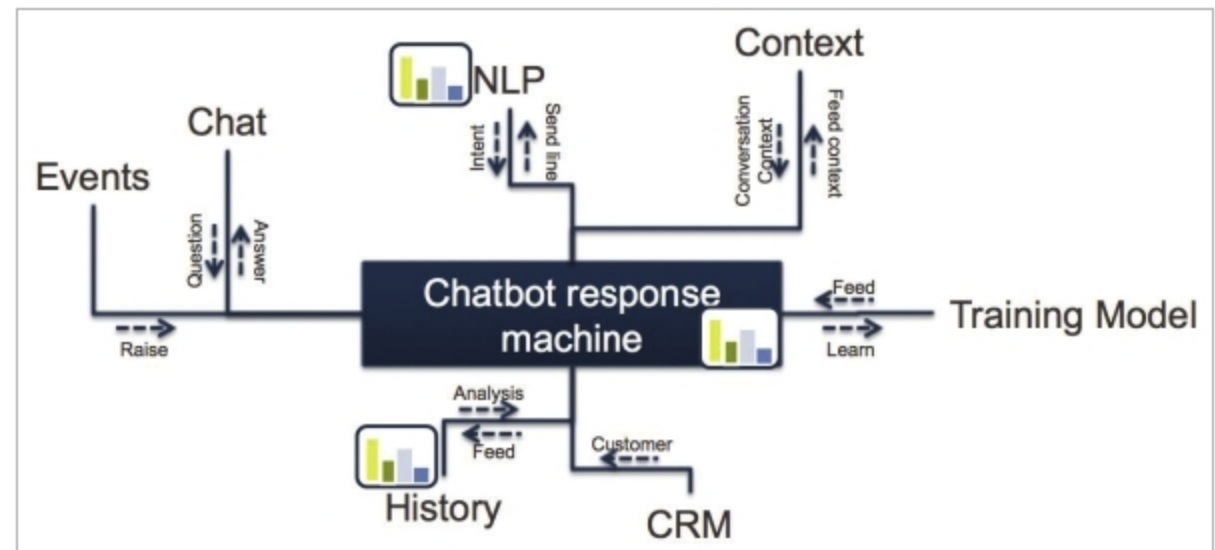


Figure 2: How a chatbot works

chatbot should be able to decide which bot ought to interact with end users using intelligent algorithms if an organisation has multiple bot based applications.

- **GUI interface:** A simple and interactive GUI is required for a rich user experience.

Stage 3 - Test: Like any other software, the bot application requires in-depth testing so that all errors or bugs can be identified and removed, making it operate intelligently in real-time applications. To test the bots, developers can make use of any of the following methodologies.

- **Using an emulator:** An emulator is a testing technique to test bots locally. The bot framework emulator is a standalone app that not only provides a chat interface, but also debugging and interrogation tools to test the bot's functionalities. Many bots can be tested simultaneously, using the emulator, during the development stages.
- **Web based testing:** Developers can make use of the Web to test the bot's functionalities. The Web based chat is the best interface to interact with and understand the functionalities of the bot, test the

live code, detect any unusual activity and then debug it.

Stage 4 - Publish: After testing, when the chatbot is ready for launch, it can be published using an online website or a data centre. Traffic can then get directed to the chatbot when an end user opens the website.

Stage 5 - Connect: It is quite important to connect the chatbot to various online apps like Skype, Facebook, WeChat, etc, to send and receive messages from all the different platforms.

Stage 6 - Evaluate: With the data collected over a certain time period after implementation, the developers can detect any errors, and use feedback to improve both performance and capabilities. With analytics, developers can get conversational-level reporting, message and channel data for further development.

Chatbots: Working, training and workflow methodologies

Chatbots work faster than normal human beings. An end user can ask a question to a chatbot that is nourished with a huge knowledge base and conversational logs, from which it can immediately understand the requirement of the user and answer the queries.